NCSS 70 YEARS OF SCIENCE

# PULSE QUEST



VOL. 03

ISSUE 01

EXPLORING THE FRONTIERS OF SCIENCE:
INNOVATIONS AND DISCOVERIES

AN INITIATIVE BY
NALANDA COLLEGE SCIENCE SOCIETY

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# Our Story.

Nalanda College Science Society is one of the most prominent and active societies in the school one largest society in our school having over 300 members in both 12-13 sections. It was initiated on 17th July 1956 and there upon have been rendering a great service to the school, country and to the world as well. Many renowned scientists, innovators, entrepreneurs, leaders and wide range of professionals have quenched their thirst of knowledge from the golden fountain of NCSS.

Nalanda College Science Society has acquired many victories in the national level as well as the international level. Rakitha Malewana, the former president of NCSS (2012-2013) can be considered as a great example. His research project on medication for HIV AIDS virus received a gold medal at the Young Scientist's Olympiad competition. Furthermore grade 13 student, Gayanga Chathushan Kuruppu won a silver medal at the I-SWEEEP Olympiad held in Texas,USA for developing an Android application to detect early warning signs of health deterioration. Also, student Rakindu Wickramarathne won a silver medal at the Young Scientist Conference held at Germany for his research.

Nalanda College Science Society had been a foster mother for such curious minds. Throughout their student career she will stand strongly illuminating mother Nalanda.

To commemorate the 100th anniversary of Nalanda College, the Nalanda College Science Society is thrilled to announce the launch of a monthly e-magazine dedicated to the wonders of science. This publication will explore cutting-edge research, spotlight pioneering discoveries, and share insights from the vibrant scientific community. Our goal is to ignite curiosity, inspire innovation, and keep readers informed on the latest advancements across various scientific disciplines. Join us as we celebrate a century of academic excellence with a publication designed to educate, engage, and excite readers of all backgrounds.

# The Spark of Pulse Quest



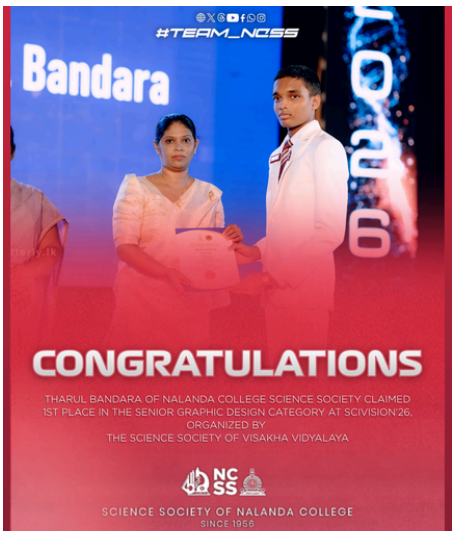
The Science Society of Nalanda College, with a proud history of 70 years, has always sought to innovate and inspire. In continuation of this tradition, launched an e-magazine called "Pulse Quest" in November 2024, conceptualized by the creative vision of students of the Science Society.

This magazine is a monthly way of bringing together students from different schools who share a passion for science and motivating them to present their talents and ideas. With "Pulse Quest", we try to present their work while infusing curiosity and teamwork among young minds.

Looking ahead, we are determined to make this platform even better in the future, with a diverse content and exciting opportunities. We extend our heartfelt gratitude to everyone who has supported us on this journey and invite you to join us in making this endeavor a remarkable success.

# NCSS Spotlight












# The Legacy Continues.



**Prof. Rahula Anura Attalage**

Rahula Anura Attalage a professor of mechanical engineering and the director of the Post Graduate Institute of the Faculty of Engineering at the University of Moratuwa in Sri Lanka. After completing college at Nalanda College Colombo, Rahula went on to earn a mechanical engineering degree from the University of Moratuwa. He obtained his master's degree in energy technology from the Asian Institute of Technology, Thailand. He also holds a PhD in energy engineering from Ecole des Mines de Paris, France. Attalage has received the University of Moratuwa's award for Outstanding Research Performances on five consecutive occasions

# Past Director Massage

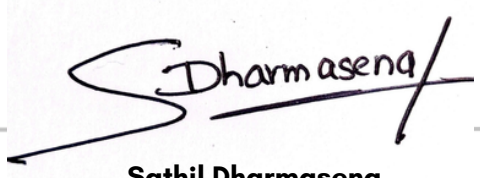
With heartfelt gratitude, I sign off as the Vice President, Director of Pulse Quest and Founder of Pulse Quest, consider it a true honour to have initiated and led one of the school science society based e-magazine platforms in Sri Lanka. To serve Mother Nalanda through a project built upon innovation, creativity, and scientific passion is a privilege I will always cherish.



Under the proud 70 year legacy of the Nalanda College Science Society, Pulse Quest grew from a simple vision into a nationwide platform uniting students who share a love for science. Through the collective dedication of our Founding Pulse Quest Team, we successfully completed 10 issues of Volume 1 and 5 issues of Volume 2, creating opportunities for young minds across Sri Lanka to share knowledge, ideas, and creativity.

A special word of gratitude goes to our Pulse Quest Team for your commitment, teamwork, and belief in this journey, which transformed a vision into a lasting reality. I also extend my sincere appreciation to the Computer, Robotics, Bio Science, Mathematics, and Aeronautical societies for joining hands with us and contributing towards making Volume 2 an extraordinary and impactful chapter of Pulse Quest. Your support, collaboration, and enthusiasm greatly strengthened this initiative and reflected the true spirit of scientific unity and innovation.

I extend my sincere thanks to the Nalanda College Science Society, our teachers, supporters, contributors, and every reader who stood with us throughout this journey. As I conclude this chapter, I do so with immense gratitude, unforgettable memories, and great pride in what we achieved together. May Pulse Quest continue to inspire, evolve, and reach even greater heights in the years to come. In the future, Pulse Quest will return with even more innovative experiences and fresh dimensions, and I extend my heartfelt best wishes to the new team as they carry this legacy forward in a way that best suits its vision and spirit.



**Sathil Dharmasena**
**Vice President**
**Founder & Director Of Pulse Quest**
**Nalanda College Science Society.**

# New Era of Pulse Quest...

# OUR TEAM

## Director
Sadew Sewmina

## Assistant Director
Tharul Bandara

## Content Manager
Nikila Yoshitha

## Editor In Chief
Sandaru Sankalpa

## Design & Feature Editor
Senuda Hansira

## Layout & Complication Editor
Pamu Disara

# Present Director Massage

It is both an honor and a profound responsibility to step into the role of Director of Pulse Quest. Standing on the foundation built by our predecessors, I take on this journey with deep gratitude for the vision, dedication, and leadership that established Pulse Quest as a pioneering e-magazine platform in Sri Lanka under the prestigious 70-year legacy of the Nalanda College Science Society.



Over the past volumes, Pulse Quest has proven to be more than just a publication—it has become a national stage for young minds to connect, innovate, and express their passion for science. As we open this new chapter, my vision is to build upon this strong heritage, introducing fresh dimensions, broader collaborations, and even greater opportunities for students across the island to share their scientific creativity.

To our incoming Pulse Quest Team, I am excited to work alongside each of you. Our success will rest on our collective curiosity, teamwork, and commitment to excellence. Together with the continued collaboration of our fellow societies, teachers, and supporters, I am confident that we will take this platform to extraordinary new heights.

To our readers and contributors, thank you for your unwavering enthusiasm and support. We are eager to bring you an enriched, inspiring experience in the upcoming issues.
May Pulse Quest continue to illuminate minds, foster innovation, and serve Mother Nalanda with pride.



**Sadew Sewmina**
**Magazine Director**
**20262027 Director Of Pulse Quest**
**Nalanda College Science Society.**

# Articles

# The Illusion of Certainty: Unmasking Forensic Evidence

In the modern courtroom, forensic evidence is often treated as the ultimate oracle. Pop culture and the infamous "CSI Effect" have conditioned juries and the public to believe that a single strand of hair, a fingerprint, or a drop of blood is an absolute guarantee of guilt.

But behind the sterile, closed doors of modern laboratories, the reality is far more fragile. Is forensic science truly infallible, or is the legal system placing blind faith in a flawed concept?

Consider DNA, universally praised as the gold standard of criminal investigations. While the genetic code itself is absolute, the interpretation of it is not.

Modern swab tests have become so hyper-sensitive that they can detect "secondary DNA transfer." This terrifying reality means an innocent person's DNA can end up on a murder weapon they never even touched, simply because they shook hands with the actual killer hours earlier, or touched a doorknob at a shared location. Science identifies the presence of DNA, but it cannot always definitively explain how or when it arrived at the crime scene.

Similarly, fingerprint analysis long considered foolproof relies heavily on human judgment rather than pure mathematics. It is not just algorithms matching exact patterns; it involves human analysts who are naturally susceptible to "cognitive bias." If an analyst is informed that a suspect has a highly violent history, their brain may subconsciously force a pattern match where none actually exists.

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Other historically accepted methods, such as bite-mark comparison and microscopic hair analysis, have recently been severely discredited by scientific reviews, leading to the exoneration of dozens of wrongfully convicted individuals who spent decades behind bars.

Furthermore, as we crash into the era of deepfakes and AI-generated realities, digital forensics faces an unprecedented crisis. When a video, an audio recording, or a digital photograph can be perfectly fabricated to frame an innocent party, what truly constitutes "hard evidence" anymore? The justice system is struggling to keep pace with technology that can seamlessly manipulate reality.

Ultimately, forensic science is a powerful tool, not a crystal ball. It is an evolving, imperfect discipline governed by human beings, and humans inevitably make mistakes. For the modern legal system to be truly just, we must strip away the illusion of forensic perfection. We must constantly question the science, rigorously challenge the methods in court, and remember that in the pursuit of justice, blind faith is the greatest enemy of truth.

-END-

NIKILA YOSHITHA
GRADE 13
NALANDA COLLEGE
COLOMBO

# REWRITING LIFE:

# THE STUNNING RISE OF GENETIC ENGINEERING

Forget dusty textbooks and static models. The "Bio Tech revolution", driven by the transformative power of genetic engineering is not merely reshaping life it's actually becoming part of the warp & weft of our future, threads of ingenuity weaving their way through medicine farming & even our understanding of what it means to be human This is not evolution, this is not creeping forward gradually - this is shining, speeding, intermingling of technology & life, promising Solutions to some of humankind's most intractable knotty problems while presenting us with fundamental ethical dilemmas.

## BEYOND CURE ENGINEERING HEALTH IN THE 21ST CENTURY

The traditional approach to medicine which seeks to cure symptoms whenever they arise, is being revolutionized by genetic engineering's precision. A new era of preventative, personalized health is dawning, in which our personal genetic code is a blue-print for health.

- **The CRISPR Precision Strike for Eradicating disease :**

Imagine having a molecular scalpel that will eliminate faulty genes causing genetic disease passed along through heredity. Game-changing gene - edited technology, CRISPR-Cas9, is making that possible From targeting cystic fibrosis & Huntington's disease at their roots to building new treatments for Sickle cell anemia & muscular dystrophy, CRISPR IS offering a future when genetic predisposition does not control fate anymore. This isn't treatment for Symptoms, this is actually eliminating disease at its root by rewriting genetic code

- **The Body's New allies Engineered Immunotherapy:**

Immunotherapy is pioneering a new age in cancer therapy.
Genetic Engineering holds out hope for turbo charging our own immune cells in order for them to spot & destroy cancer. CAR T-cell therapy is a case in point, genetically modifying a patient's T-cells to generate chimeric antigen receptors (CARS) specifically for targeting cancer cells alone. This customized treatment, leveraging the body 's own immune system is making history in treating previously untreatable cancers.

- **Biologics & Beyond: Living drug factories :**

Genetically altered organisms from bacteria to mammalian cells are being turned into high-tech bio-factories for producing complex therapeutic proteins known as Biologics. Diabetes medicines such as Insulin, Monoclonal antibodies for autoimmune disease & cancer, and an expanding list of next generation drugs all are products of this bio-manufacturing era. Scientists already are exploring methods for making "living medicines - genetically engineered bacteria that can inhabit living tissue & deliver therapeutic payloads directly into diseased tissue.

- **The potential for our Genes to Predict:**

Prevention for all people: Our genetic code itself contains great amounts of information about our predisposition toward numerous diseases. The genetic age is pushing forward development of the new diagnostic technology that will be capable of recognizing these predispositions early, which will allow us to take preventative measures based upon our own genetic code. This personalized prevention pathway has potential for changing healthcare provision from a reactive system into a preventative system.

**BUILDING RESILIENCE:**

Reimagining our relationship with nature & food Genetic Modification is not solely for curing people but also for addressing global issues related to food security & sustainability.

- **Climate Resilient Crops through Engineering:**

With our world grappling with the worsening impacts of climate change, genetic engineering is one of the most effective ways of producing crops that are tolerant to heat, drought & high salinity. Having heat tolerance & drought tolerance engineered into crops IS crucial in addressing food security for today's fast-changing world. This goes beyond yield enhancement; it's about adding resilience into our food system.

- **Nutritional Fortification:**

Bioengineering a Healthier Diet: Malnutrition, & in particular malnutrition in developing countries, is one of the leading global health priorities. With genetic engineering, we can fortify staple foods with improved nutrition. "Golden Rice", genetically engineered for beta-carotene Can antecedent to Vitamin A), offers a wonderful example of how bio-engineering can be put into practice in order to counteract deficiencies in vitamins. Scientists are currently seeking to enhance levels of essential amino acids, iron & other essential nutrients in foods widely consumed.

- **Sustainable Bio-factories:**

Engineering for a Greener World: Principles in genetics are applied in order to push innovations toward cleaner industrial processes. Micro-organisms can be genetically engineered for maximum production from renewable resources into biofuels, offering cleaner alternatives compared to fossil fuel use. Similarly, scientists are genetically engineering microbes for plastic waste degradation & the production of biodegradable commodities for reducing growing plastic pollution concerns. This "Bio-economy" holds tremendous potential for reducing our footprint in the environment.

## PRECISION AGRICULTURE

- **Data-driven crop Enhancement**

Genetic modification & data analysis are being combined to bring about a new era in precision farming. With knowledge about the genetic makeup of plants & how those traits would react under specific environmental conditions, Scientists can develop highly targeted genetic adjustments that optimize resource utilization & minimize environmental impact Data has the potential to make farming processes sustainable & cost effective

## THE MORAL FRONTIER

Exploring the unknown expanse of life's code the immense potential for genetic engineering necessarily raises fundamental ethical & social questions which need to be cosidered & debated openly

- **The Germline Divide Editing humanity's future?**

Perhaps the most Contested ethical discussion is about germline editing - making permanent changes to sperm, egg or embryo DNA that would be passed down through generations. Promising an end to genetic disease, this raises questions about possible unforeseen long-term consequen -ces & creating "designer babies" genetically altered for non-medical traits which would perpetuate social inequalities.

- **Ecological Ripple Effects:**

Surprising Consequences in Nature Release into the environment of genetically modified organisms has raised concerns about potential ecological impacts, that is, horizontal gene transfer to relatives in nature & unexpected impacts on diversity. Appropriate risk assessment & careful regulation need to be established in order to prevent such potential harms from happening.

- **Equity & Access :**

Guaranteeing a just share in the gains With advancing genetic engineering technology, we need to ensure equitable access to their gains. Will these therapies that run lives around & enhanced crops benefit all humans, or will they exacerbate existing Social & economic divides? Questions have to be confronted about affordability & access.

- **A new definition of "Natural":**

The shifting frontiers of Life: Genetic modification puts into question our traditional concept of what is "Natural & forces us to consider fundamental questions about our own agency in determining the destiny of life here on earth. Open discussion between scientists, ethicists, decision-makers & Citizens is required in addressing these complex philosophical & societal concerns & in creating responsible standards for working with this powerful technology.

In conclusion, Genetic Technology is no longer a distant prospect but an evolving reality that is reshaping life's script. From healing debilitating disorders to creating robust food systems & building a tomorrow that is sustainable its capacity for all humanity is immense. But all such potential comes along with a need for wisdom. While we continue to unravel the code of life & continue perfecting our art at code-manipulation, a thoughtful & ethical reaction is called for. By accepting change & examining the social & environmental consequences cautiously, we can see that the BioTech Revolution" weaves a future in which genetic engineering's strands are interwoven in the well-being of all life upon our earth.

NETHMI MEDERIGAMA
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ST. JOSEPH'S BALIKA VIDYALAYA, KEGALLE

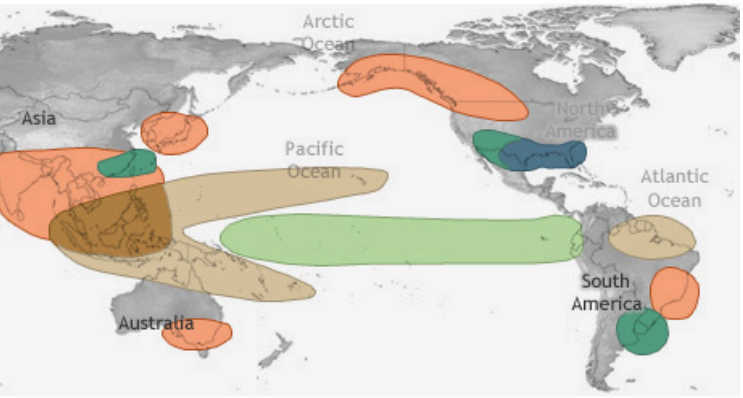
# El Nino:

# La Nina Phenomena

El Niño and La Niña are opposite phases of a naturally occurring climate pattern known as the El Niño-Southern Oscillation (ENSO), involving fluctuating ocean temperatures in the central and eastern equatorial Pacific, coupled with changes in the overlying atmosphere.

## Overview

ENSO has two opposite extreme phases: the warm phase, known as El Niño, and the cold phase, known as La Niña. Between these extremes lies a third state called ENSO-neutral. El Niño can typically last up to 18 months and La Niña up to three years.




El Niño is characterized by unusually warm ocean surface temperatures in the central and eastern Pacific. The name originates with Peruvian fishermen and sailors who noticed a warm current, associated with reduced fish catches, and named it El Niño ("the Christ Child"), likely because it typically became noticeable around Christmas . In contrast, a La Niña event is characterized by unusually cool ocean surface temperatures in the same region and an intensification of the prevailing east to west surface winds.

In the Pacific Ocean, warmer-than-normal ocean temperatures usually mean El Niño conditions, which happen when easterly trade winds become weaker. Cooler-than-normal temperatures usually mean La Niña conditions, when easterly winds become stronger. During the Neutral phase, ocean temperatures in the tropical Pacific stay close to normal.

The transition between these extreme phases typically occurs every two to seven years. Both El Niño and La Niña generally begin developing between March and June and reach their peak intensity between November and February.

The effects of each El Niño/La Niña event vary depending on the intensity, duration, time of year when it develops, and how it interacts with other climate variability modes (such as the Indian Ocean Dipole). Not all regions of the world are affected, and even within a region, impacts can be different. Even when ENSO is neutral, extreme weather can still occur.

The strength of an ENSO event is highly significant – whether it is classed as weak, moderate, strong or very strong. WMO does not use the term Super El Niño or La Niña.

Improved seasonal forecast models can now predict climate patterns one to six months in advance. By identifying trends in heavy rainfall (potentially leading to floods) and drought conditions, this information provides early warning signals that support planning in climate-sensitive sectors such as agriculture and health, while also informing the activation of government emergency protocols and the pre-positioning of humanitarian supplies. Together, these actions help to protect resources and save hundreds of millions of dollars and many thousands of lives.

A multiyear La Niña lasted from late 2020 to early 2023, followed by the strong El Niño of 2023–2024. ENSO-neutral conditions began in April–June 2024 and generally prevailed until September–November 2025, with a brief dip into weak La Niña conditions during late 2024 and early 2025. La Niña developed again in late 2025 and lasted until early 2026.

WMO's Global Seasonal Climate Update issued in May 2026 signaled a clear shift in the Equatorial Pacific: sea-surface temperatures are rising rapidly, pointing to a likely return of El Niño conditions as early as June–July–August 2026. Forecasts indicate there is a "nearly global dominance of above-normal land surface temperatures" in the upcoming three-month period, and regional variations in rainfall patterns.

WMO El Niño/La Niña Update issued in June 2026 confirmed an 80% likelihood of an El Niño event through June–August 2026. Probabilities for this to continue until at least November are near or above 90%. Although some uncertainty remains about El Niño peak strength and timing, most forecast models suggest it will be at least moderate and possibly strong.

WMO's monthly WMO Global Seasonal Climate Update issued in July 2026 indicates a rapid development into a strong El Niño event during July to September 2026. Multi-model ensemble forecasts from leading global producing centres indicate a consistent and significant warming of ocean temperatures across the central and eastern equatorial Pacific, with seasonal-average sea-surface temperature anomalies expected to exceed 2°C in key monitoring regions.

There is no evidence that climate change increases the frequency or intensity of El Nino events. It can amplify associated impacts because a warmer ocean and atmosphere increase the availability of energy and moisture for extreme weather events such as heatwaves and heavy rainfall.

## Impact

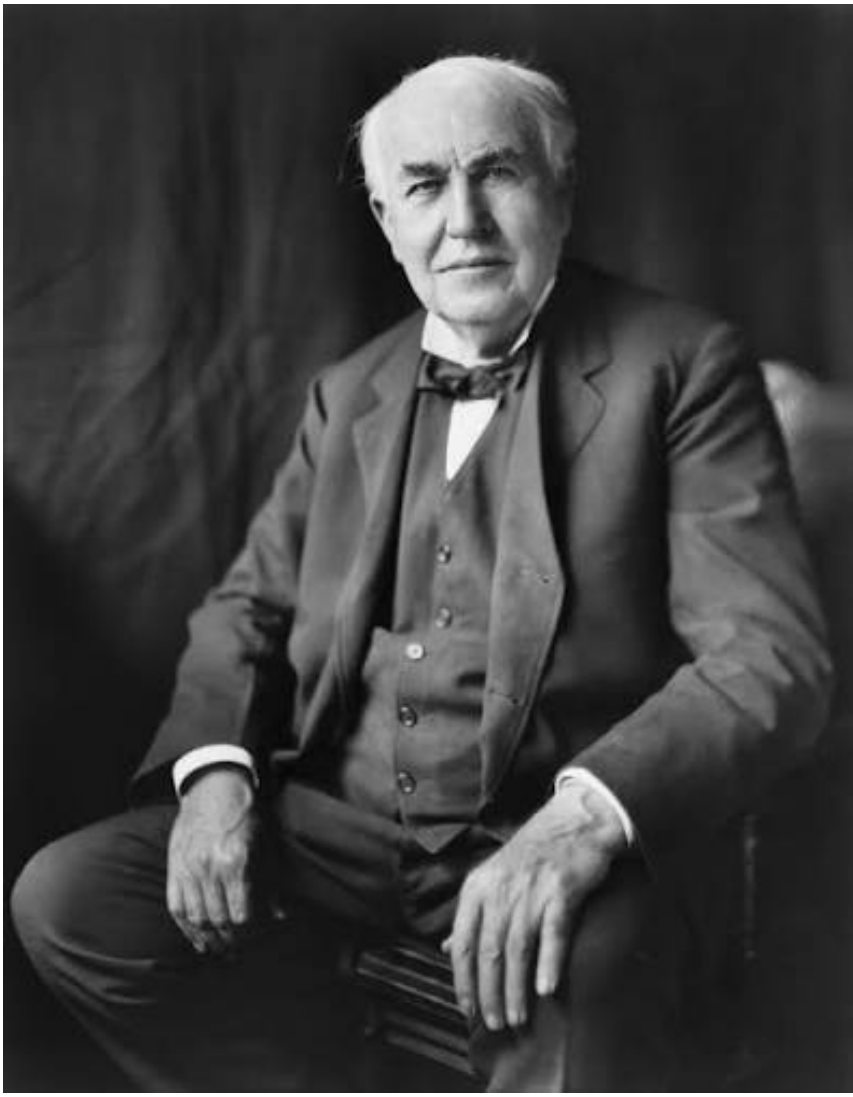
ENSO is a major influencer of climate and weather patterns. El Niño events affect temperature and rainfall patterns in different regions and typically have a warming effect on the global climate - the most pronounced impacts are in the year following development. Thus, 2024 was the hottest year on record because of the combination of the powerful 2023–2024 El Niño and human-induced climate change from greenhouse gases.

El Niño is associated with increased rainfall and flooding in parts of South America, East Africa, and the southern United States; drought conditions in eastern and northern Australia, Indonesia, southern Africa, and parts of South Asia due to suppressed monsoon activity; and reduced Atlantic hurricane activity.

The timing and intensity can make a difference. Thus, El Niño is associated with increased rainfall over East Africa, often resulting in a wetter-than-normal, and sometimes severe, “short rains” season between October and December. It is also typically linked to suppressed rainfall during the June-September season over the northern parts of the region, including Sudan, South Sudan, parts of Ethiopia and western Kenya.

## Let's learn about Thomas Alva Edison (1847-1931),

# The inventor of the light bulb.



Thomas Alva Edison was born on February 11, 1847, in the American village of Milan. When he was 10 years old, he set up his own laboratory in the basement of his house. In 1860, he worked in the telegraph office in the United States and Canada.

In 1876, he sold his telegraph printing machine. With the money he earned, he set up his own laboratory in Menye Park, New York, where he developed the carbon resistance transmitter and the gramophone , the world's first machine that could record and reproduce the human voice.

In 1879, he made another great invention. It was the electric bulb that we all use today. In 1880, he lit up Moklo Park for the first time with 100 electric bulbs. This was just one of his discoveries. The first city in the world to be illuminated by electric bulbs was New York City .

That glorious day is recorded as September 4, 1882. That night, 75 kilometers of cables were laid, providing electricity to over 1,000 homes in the area for the first time.

In 1887, he built a new science laboratory in West Orange County. In this laboratory, he developed the first motion picture camera for still photography, the Kinetograph. This camera used Eastman film. After cleaning it, people came to see it in amazement. People considered Edison a technological genius.

In 1889, a great international exhibition was held in Paris. The most important event was the exhibition of a huge number of new scientific discoveries, all of which were the work of one man. This great genius, Thomas Alva Edison, was warmly welcomed in Paris.

He had patented over 100 inventions that claimed his inventions and rights. His determination to do so may have been influenced by his childhood. His mother, a teacher, provided him with everything he needed. As a child, he hoped to earn his own money for his research. Accordingly, he sold newspapers and sweets to passengers on the Grand Trunk Railway between Port Peoria and Detroit. In 1861, he was eager to learn about the American Civil War, so he bought an old hand-operated printing press for $12. He set up his own printing press on the train, using borrowed paper.

He named it the Grand Trunks Herald. It was the first newspaper to be printed on a train. He also set up a small laboratory on the train, where he conducted various experiments and recorded them in his spare time.

An accident while selling newspapers left him deaf in one ear. Moreover, his laboratory on the train caught fire, so the angry officers ordered him not to trade from the train. They also threw out the laboratory supplies. When he tried to sell the telegraph machine he had built through obstacles to the head of the stock exchange at the time for $3,000, the head realized its value and gave him $40,000. Edison's good times began from then on. However, he continued to engage in scientific research throughout his life. Despite his great discoveries, he was a very introspective and modest person. He often said that although 1% of this intellectual power of mine was a divine inheritance, the remaining 99% was his own sweat and tears. This great man breathed his last on October 31, 1931. In honor of him, Thomas Alva Edison's name was added to the Book of Greatest Americans in 1960.

-This article is an excerpt from the book The Greatest of the World by K.R. Aryasinghe-

# Let's examine vitamins and their importance

Vitamins are organic compounds, like carbohydrates and lipids. These vitamins contribute to biochemical reactions in the body and are very important for maintaining the health of the body. These vitamins can be classified into two main groups based on their solubility. Vitamins B and C are water-soluble, while vitamins A, D, E, and K are insoluble in water. However, these are fat-soluble.
Now let's consider these vitamins separately.



## Vitamin A

It is extremely important for the formation of visual pigments that aid in vision. It also helps maintain skin radiance and health.

D*eficiency symptoms*

1) Night blindness

2) The appearance of Bitou spots in the eye

3) Dry skin

4) Blisters like thorns around the elbows and knees

5) Occurrence of diseases related to the respiratory system

## Vitamin B

It is a component necessary for the proper functioning of nerve tissue and the formation of red blood cells. Vitamin B is also important for maintaining healthy skin, fat metabolism, bone marrow formation, the maturation of red blood cells, and the formation of antibodies.

Vitamin B is a complex vitamin. It consists of vitamins B1, B2, B3, B4....., B12. These are mainly obtained from food, and some vitamins are produced by bacteria living in the human intestine.

*Deficiency symptoms*

1) The occurrence of berry disease

2) Soreness at the corners of the mouth

3) Anemia

4) Dryness and discoloration of the skin

## Vitamin C

Another vitamin that is important for skin health. It is also important for the formation of tooth enamel and the synthesis of collagen fibers.

*Deficiency symptoms*

1) Weakening of gums
2) Internal bleeding
3) Time taken to heal
4) Scurvy

## Vitamin D

This vitamin controls the absorption of calcium and phosphorus in the body. Rickets is caused by its deficiency.

## Vitamin E

This is important for cell and tissue growth.

*Deficiency symptoms*

1) Premature birth
2) Accelerated breakdown of red blood cells
3) Defects in cell division
4) Problems related to reproduction

## Vitamin K

This vitamin is important for making components needed for blood clotting. A deficiency in this vitamin causes delayed blood clotting.

# CONCLUSION...

PULSE QUEST

Initiated by
Nalanda College Science Society